

A Complete Course in Physics (Graphs) - First Edition

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Preface and Acknowledgements

This book is a research evaluate stuff on Kinematics covering graphs. The material has got acclaim locally and now looking for it internationally.

I would like to thank my family, well wishers for their support and consistent encouragement and most of all God for it's grace and Guidance.

Chapter 1

Concepts of Graphs

1.1 The Equations of motion and the origin of Graph Handling

1.1.1 The First Equation

The Equation $v = \frac{dx}{dt}$ in linear motion implies

- i) The **Slope** of **Position-Time Graph** is **Instantaneous Velocity**.
- ii) The **Area** under the **Velocity-Time Graph** is **Change in Position**.
{ The second one requires the manipulation , $dx = vdt$ i.e. $\int dx = \int vdt$ }

The equations can be further manipulated to obtain the Speed Time Graph , where

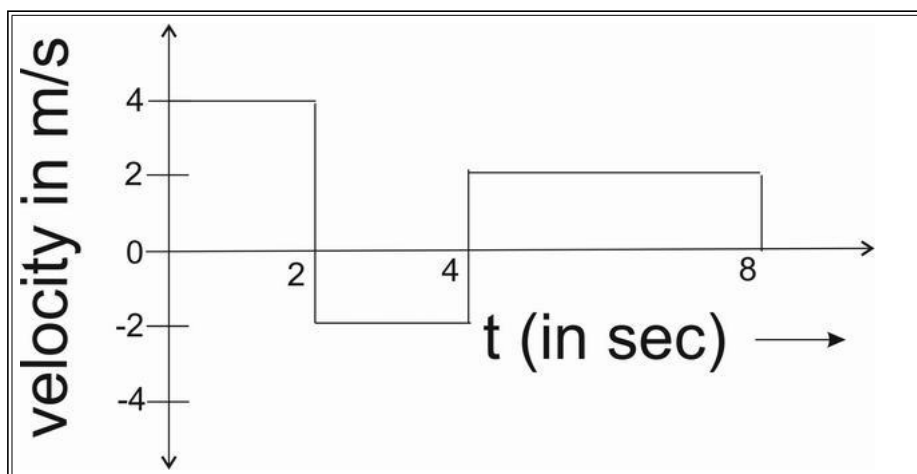
$$\text{speed} = \text{rate of change of distance wrt time}$$

Few of the following examples illustrate this concept:

Example: On a displacement-time graph, two straight lines make angles of 30° and 60° with the time-axis.
The ratio of the velocities represented by them is

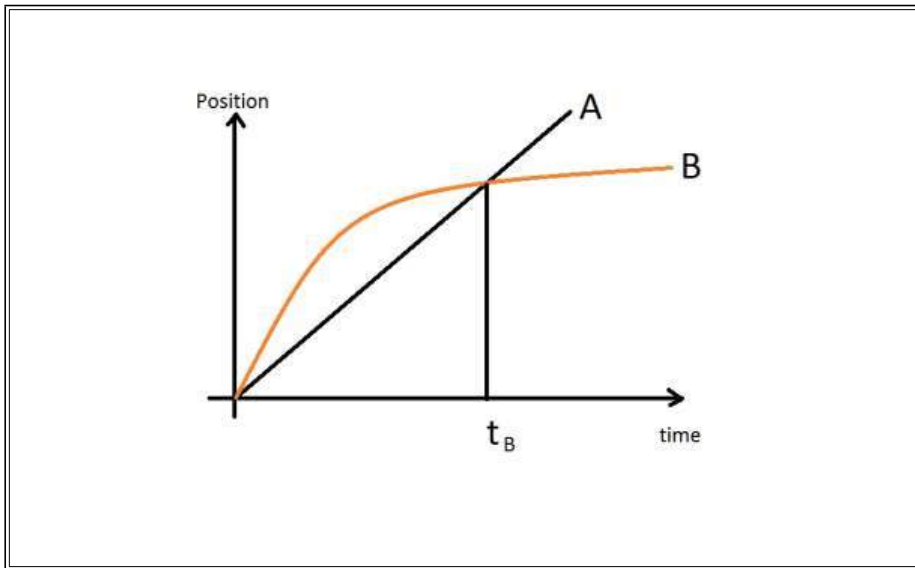
- a) $1 : \sqrt{3}$
- b) $1:3$
- c) $\sqrt{3} : 1$
- d) $3:1$

Example: A body is moving in a straight line as shown in velocity-time graph. The displacement and distance travelled by body in 8 second are respectively:



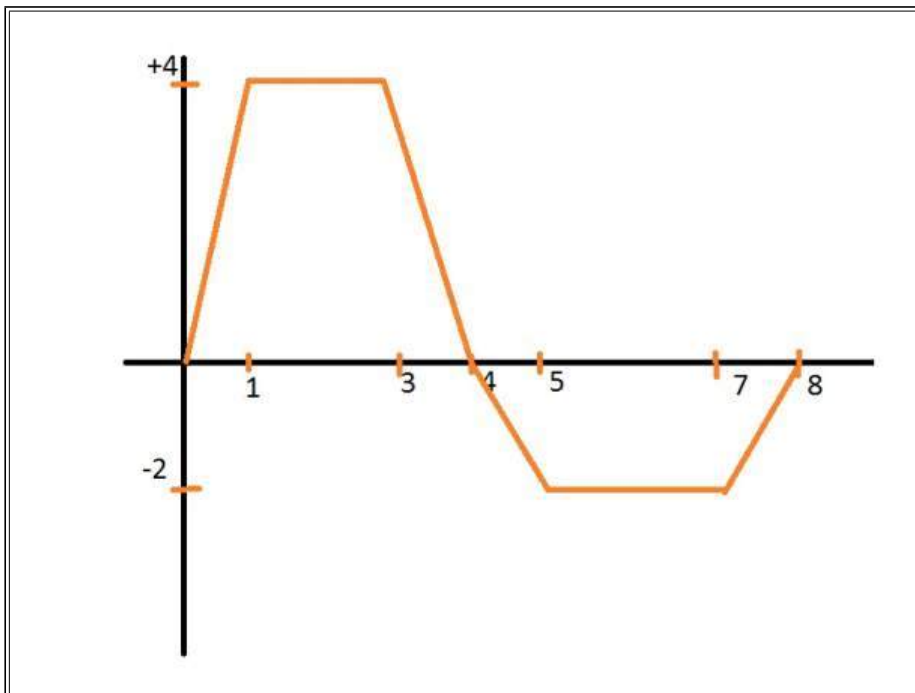
- a) 12 m, 20 m
- b) 20 m, 12 m
- c) 12 m, 12 m
- d) 20 m, 20 m

Example: The graph shows position as a function of time for two trains running on parallel tracks. Which statement is true?



- a) At time t_B both trains have the same velocity.
- b) Both trains have the same velocity at some time after t_B .
- c) Both trains have the same velocity at some time before t_B .
- d) Somewhere on the graph, both trains have the same acceleration.

Example: The velocity-time graph of a particle in linear motion is as shown. Both v and t are in SI units. The displacement of the particle is



- a) 6 m
- b) 8 m
- c) 16 m
- d) 18 m

1.1.2 The Second Equation

Proceeding similar to above, the equation $a = \frac{dv}{dt}$ implies

i) The **Slope** of **Velocity-Time Graph** is **Instantaneous Acceleration**.

ii) The **Area** under **Acceleration-Time Graph** is **Change in Velocity**.

{ The second one requires the manipulation , $dv = a dt$ i.e. $\int dv = \int a dt$ }

A few of the following examples illustrate it.

Example: A car starts from rest acquires a velocity v with uniform acceleration $2ms^{-2}$ then it comes to stop with uniform retardation $4ms^{-2}$. If the total time for which it remains in motion is 3 sec, the total distance travelled is:

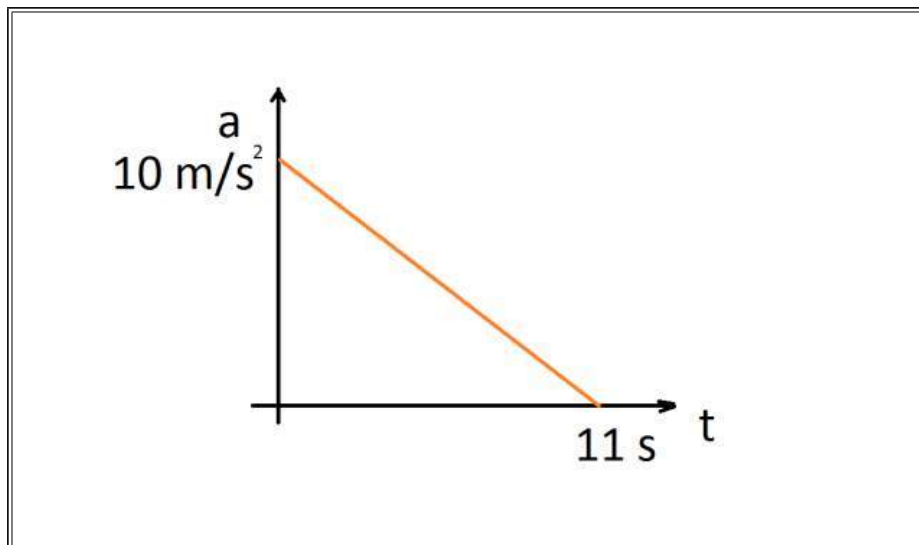
a) 2 m

b) 3 m

c) 4 m

d) 6 m

Example: A particle starts from rest. Its acceleration (a) vs time (t) is as shown in the Figure. The maximum speed of the particle will be



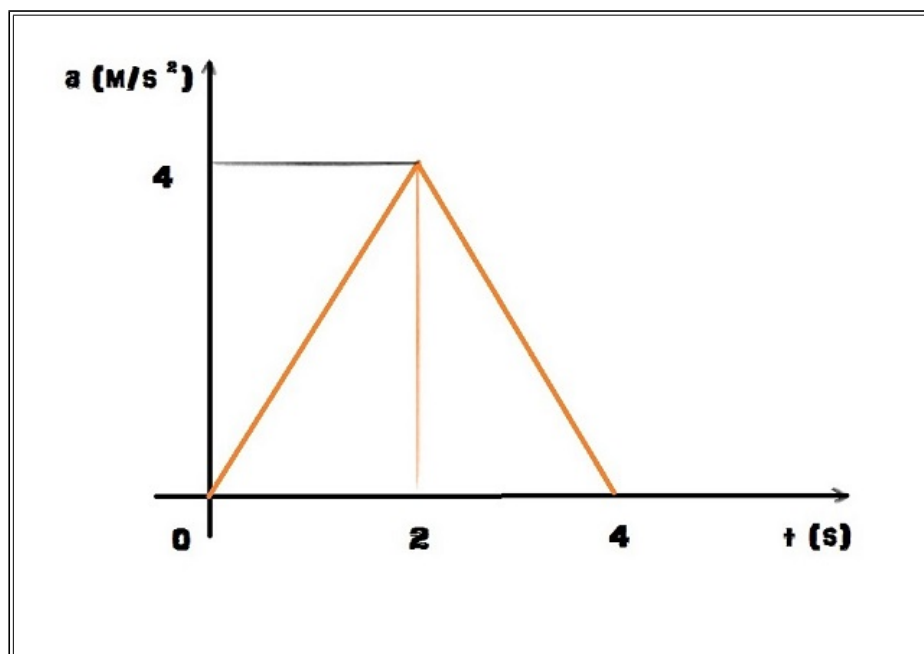
a) 110 m/s

b) 55 m/s

c) 550 m/s

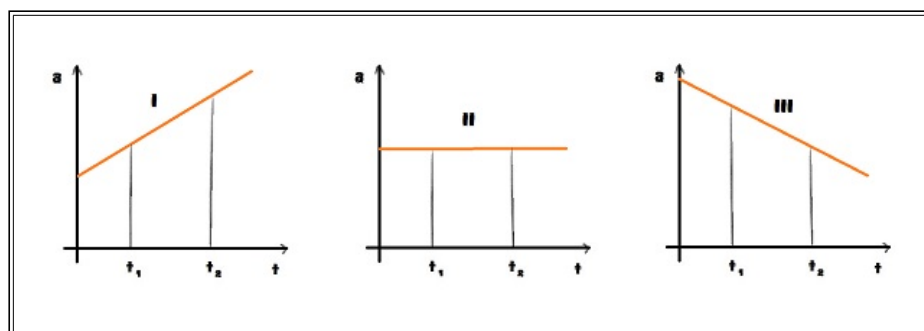
d) 660 m/s

Example: Acceleration-time graph of a particle moving in a straight line is shown in Figure. The velocity of particle at time $t = 0$ is 2 m/s. Velocity at the end of fourth second is



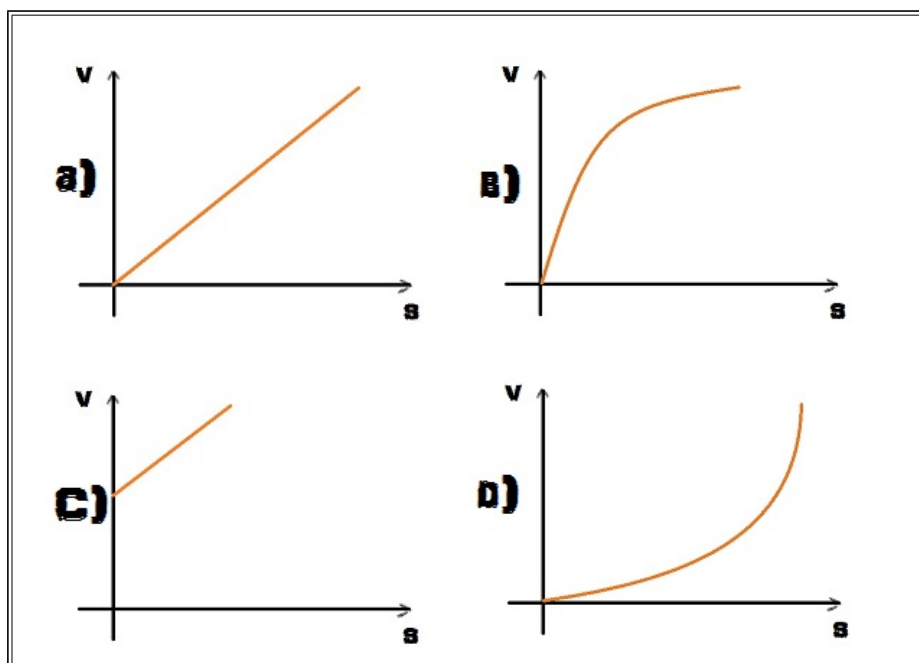
- a) 8 m/s
- b) 10 m/s
- c) 12 m/s
- d) 14 m/s

Example: Each of the three graphs represents acceleration vs time for an object that already has a positive velocity at time t_1 . Which graph/graphs show an object whose speed is increasing for the entire time interval between t_1 and t_2 ?



- a) Graph I only
- b) Graphs I and II
- c) Graphs I and III
- d) Graphs I, II and III

Example: A body starts from rest and moves along a straight line with constant acceleration. The variation of speed v with distance s is given by the graph



1.1.3 The Acceleration-Position Graph Variate

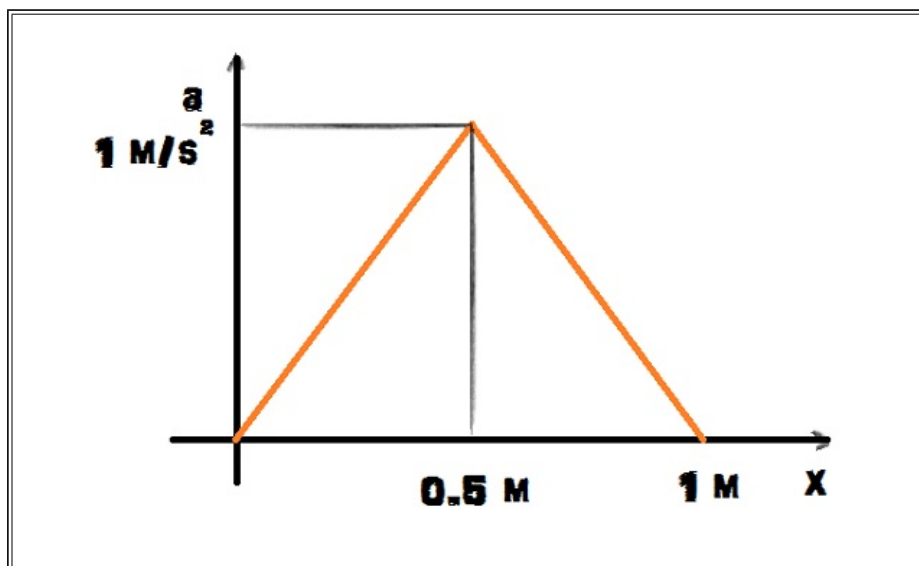
This kind of graph requires the manipulation of the Equation $a = \frac{dv}{dt}$ as follows

$$a = \frac{dv}{dx} \cdot \frac{dx}{dt}$$

$$\Rightarrow a = \frac{dv}{dx} \cdot v$$

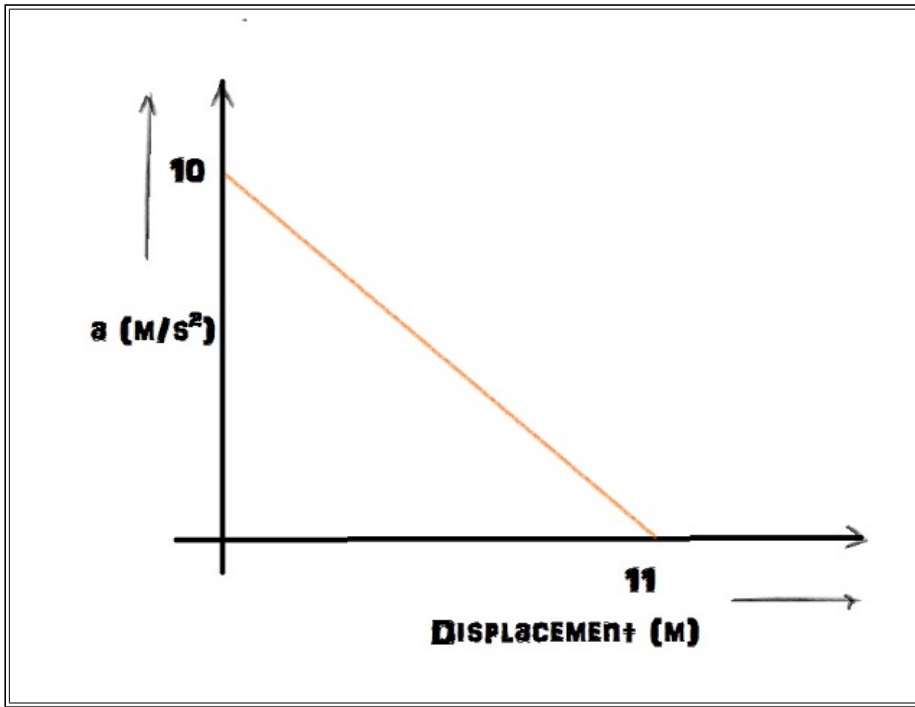
$\Rightarrow adx = vdv$ and integration can be performed to further solve it.

Example: A body, initially at rest, starts moving along x-axis in such a way that its acceleration vs displacement plot is as shown in the Figure. The maximum velocity of the particle is



- a) 1 m/s
- b) 6 m/s
- c) 2 m/s
- d) None of these

Example : A particle initially at rest, it is subjected to a non-uniform acceleration a , as shown in the gure. The maximum speed attained by the particle is



- a) 605 m/s
- b) 110 m/s
- c) 55 m/s
- d) 110 m/s

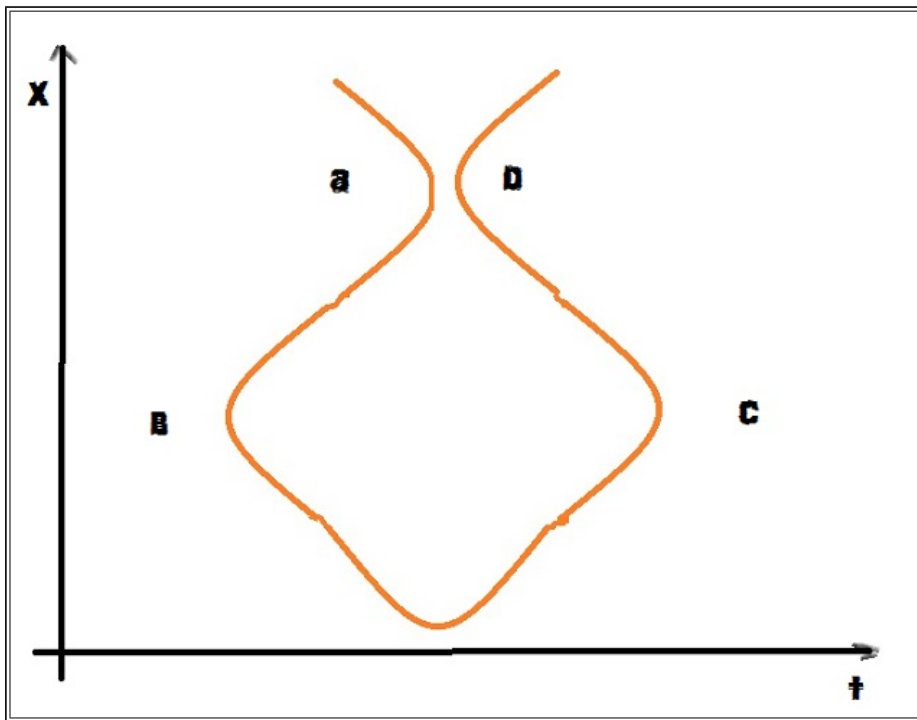
1.2 Other Types of Graphs

1.2.1 Sign of Acceleration from Position-Time Graph

The sign of Acceleration can be determined from the Position-Time Graph. The methodology involves looking at the Concavity of the Graph

- i) If the graph is Concave-Up, the Acceleration is Positive.
- ii) If the graph is Concave-Down, the Acceleration is Negative.
- iii) If the graph is a straight line, the Acceleration is ZERO. { Irrespective of any other factor , such as the slope or direction of line }

Example: The graph given below is a plot of distance vs time. For which labelled region is the “Velocity Positive and the Acceleration Negative”

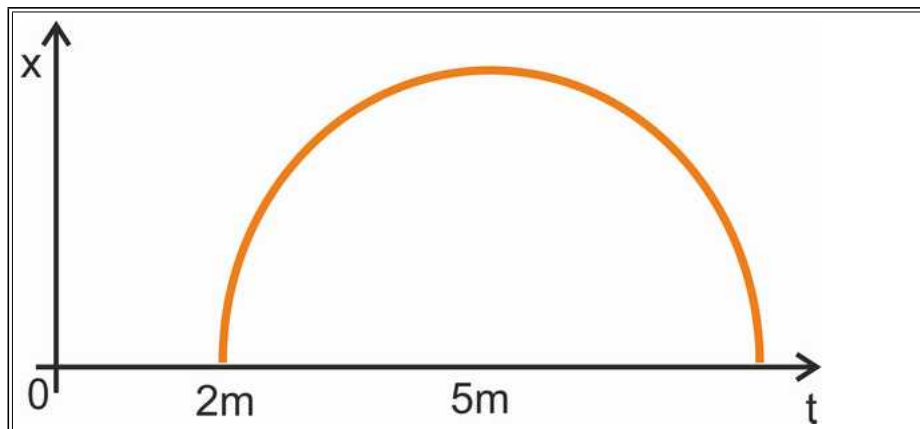


- a) a
- b) b
- c) c
- d) d

1.2.2 The Average-Velocity / Instantaneous Velocity , Equal Case

We know , that (in a x-t graph) the slope of the Secant is the Average Velocity , whereas the slope of Tangent is the Instantaneous Velocity. The point where these two lines coincide, is the point where Average Velocity is equal to Instantaneous Velocity.

Example: Position-time graph is shown which is a semicircle from $t = 2$ to $t = 8$ s. Find time t at which the instantaneous velocity is equal to average velocity over first t seconds,



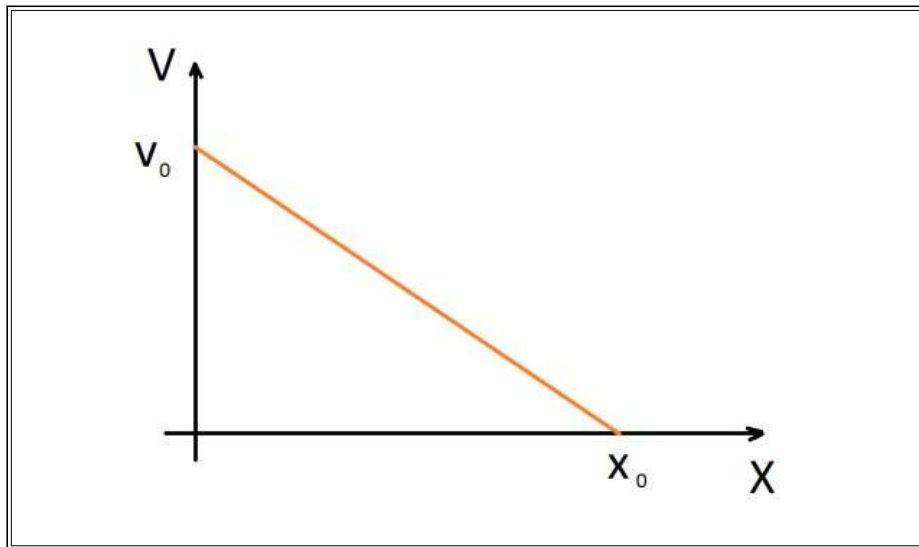
- a) 4.8 s
- b) 3.2 s
- c) 2.4 s
- d) 5 s

1.2.3 The Velocity-Displacement Case

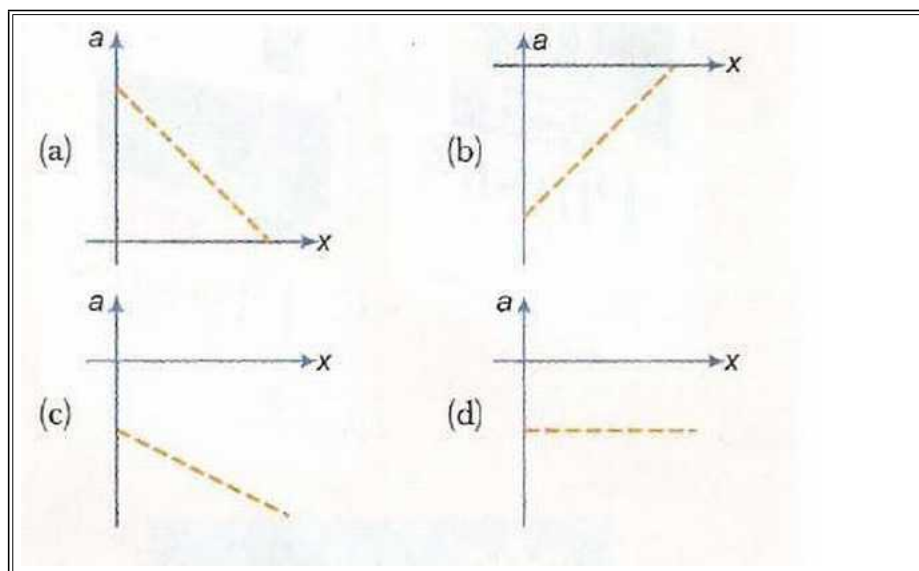
This can be handled in a similar way as Acceleration-Displacement case by integrating the respective equation. Here the problem is of $v = f(x)$ type, which can be integrated by writing $\frac{dx}{dt} = f(x)$

i.e. $dx = f(x)dt$

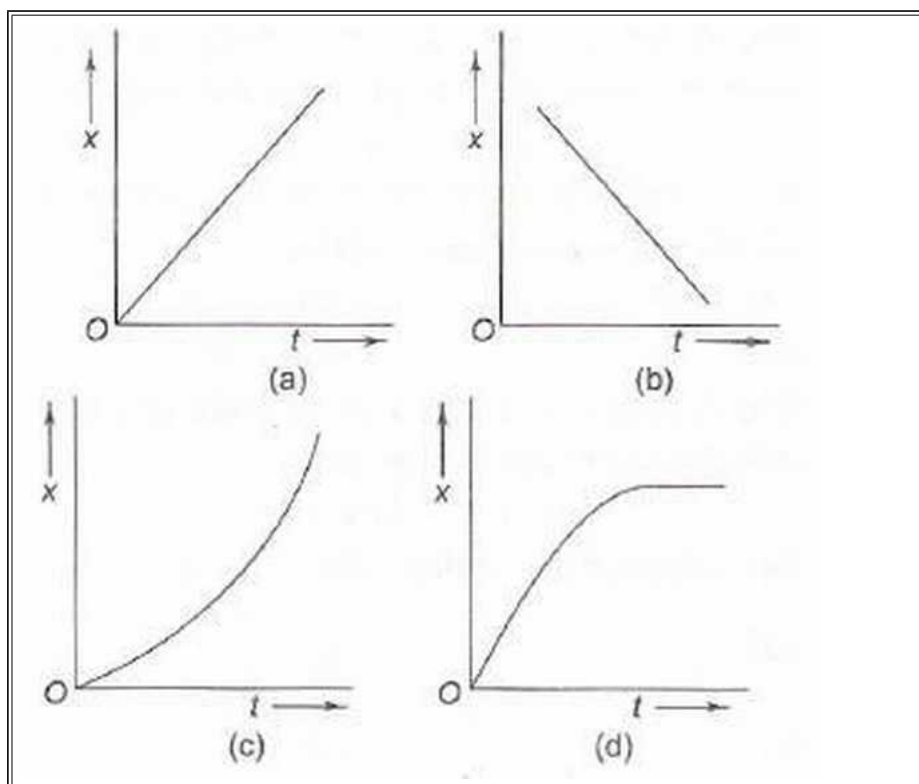
Example: The velocity-displacement graph of a particle moving along a straight line is shown here.



The most suitable acceleration-displacement graph will be



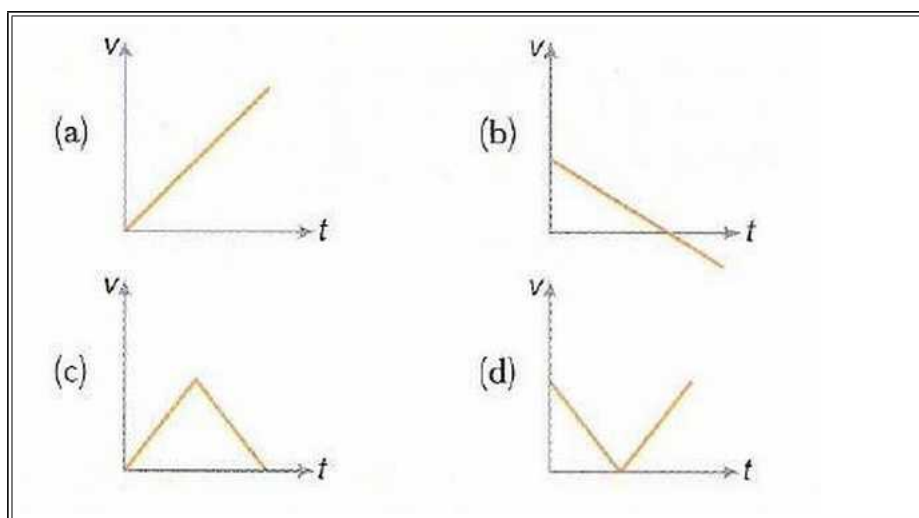
Example. The velocity (v) of a body moving along the positive x -direction varies with displacement (x) from the origin as $p v = k x$, where k is a constant. Which of the graphs shown in Fig. correctly represents the displacement-time ($x - t$) graph of the motion?



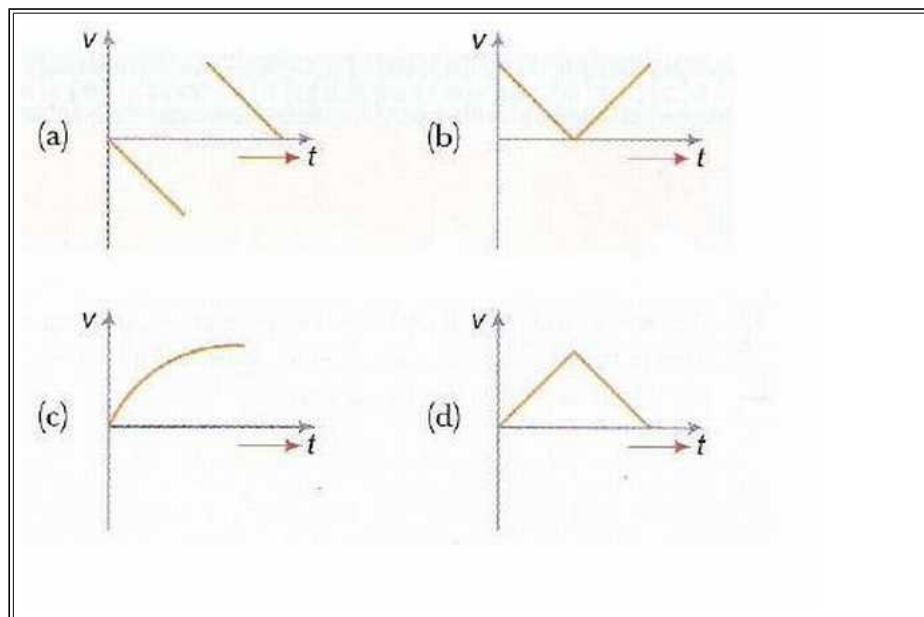
1.2.4 Motion Under Free Fall due to Gravity

In such examples, the governing equations rule and the coordinate system needs to be properly chosen.

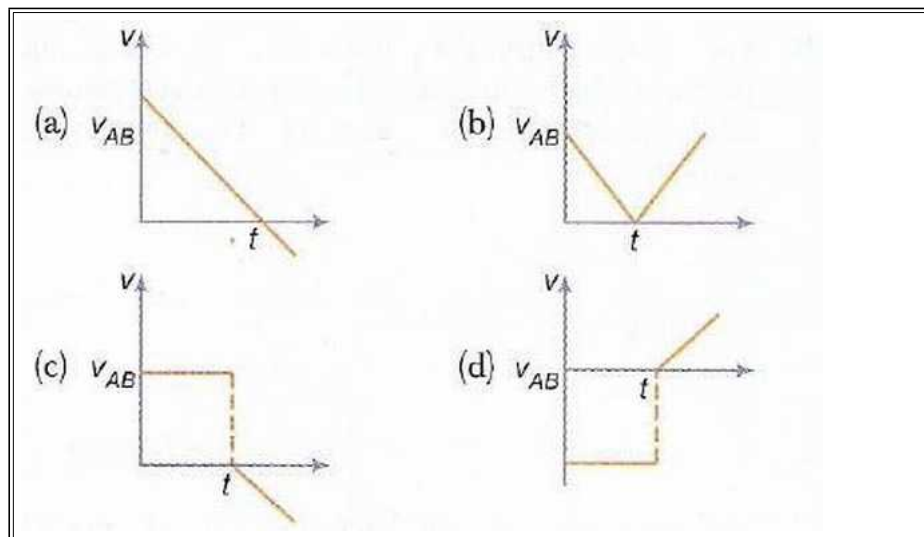
Example: Which of the following graphs correctly represents velocity-time relationship for a particle released from rest to fall freely under gravity?



Example: The velocity-time graph of a body falling from rest under gravity and rebounding from a solid surface is represented by which of the following graphs?

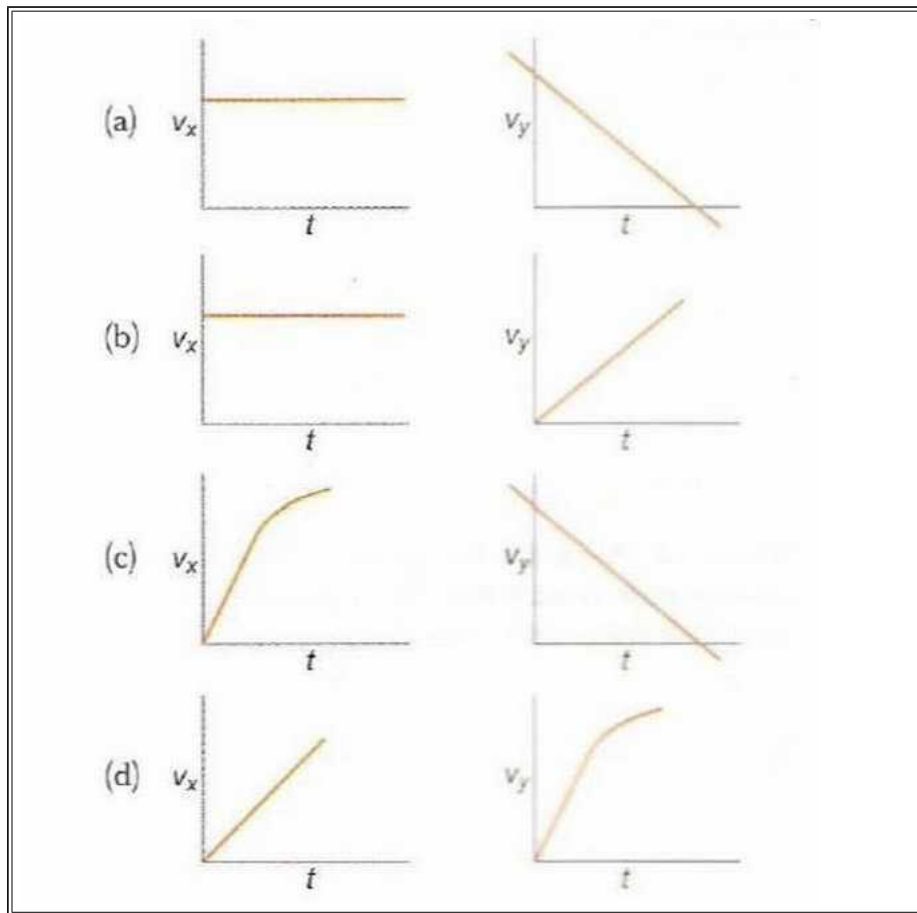


Example: A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of h . Simultaneously another body B is dropped from height h . It strikes the ground and doesn't rebound. The velocity of A relative to B vs time graph is best represented by (upward direction is positive.)



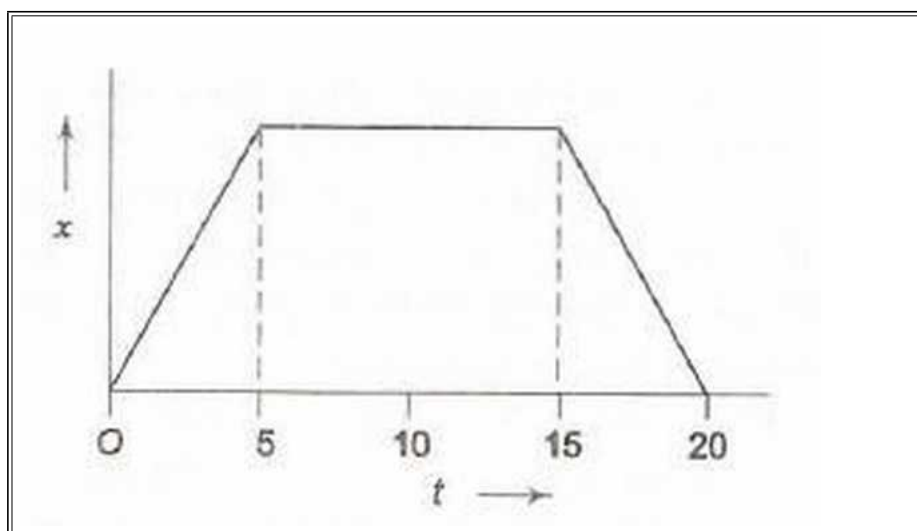
1.2.5 Projectile Motion

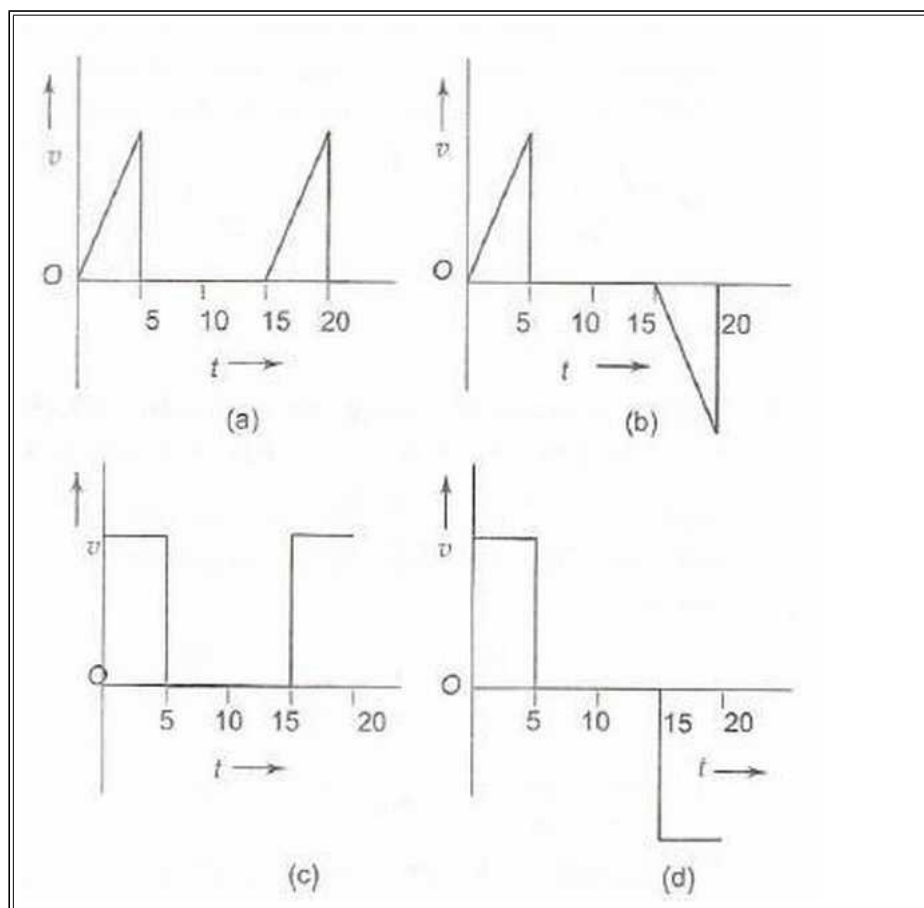
Example: A shell is fired from a gun at an angle to the horizontal. Graphs are drawn for its horizontal component of velocity v_x and its vertical component of velocity v_y .



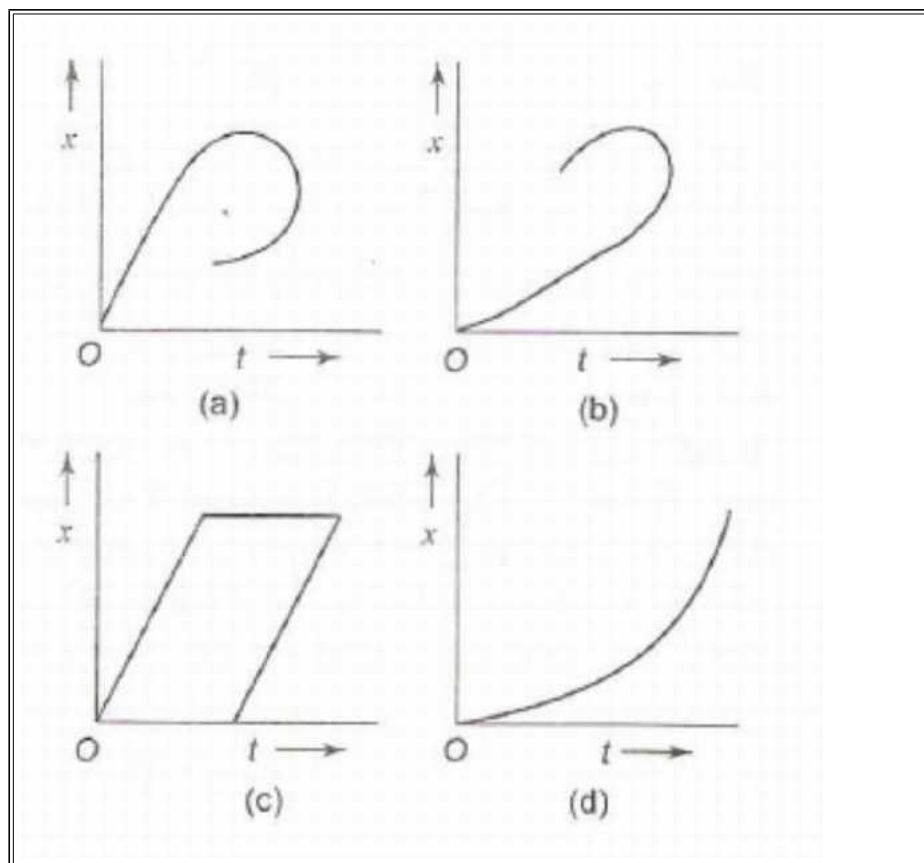
1.2.6 Miscellaneous

Example. Figure shows the displacement-time ($x-t$) graph of body moving in a straight line. Which one of the graphs shown in Fig. represents the velocity-time ($v-t$) graph of the motion of the body.

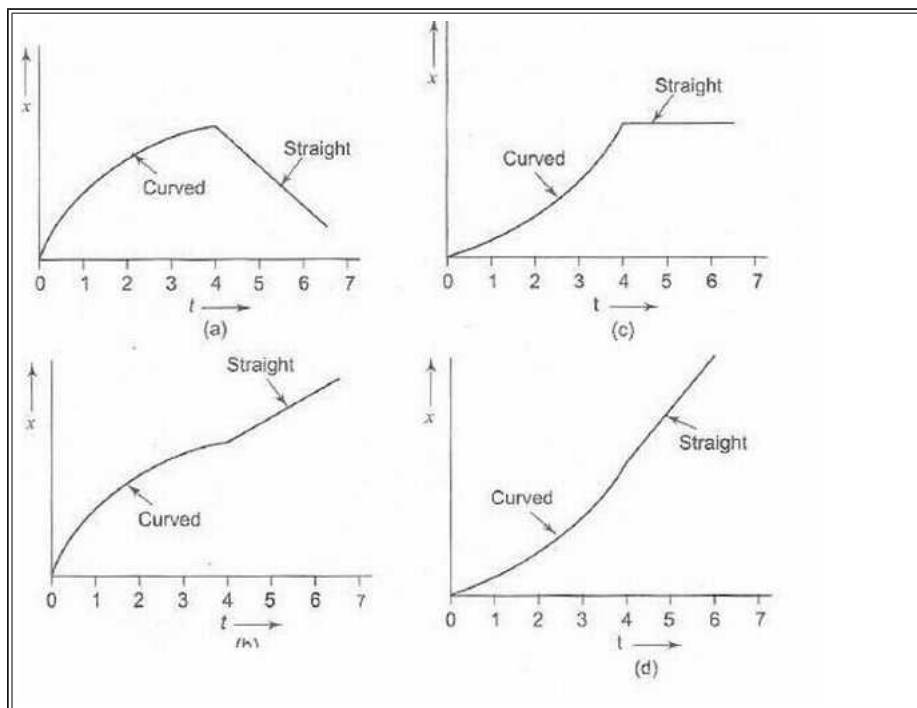




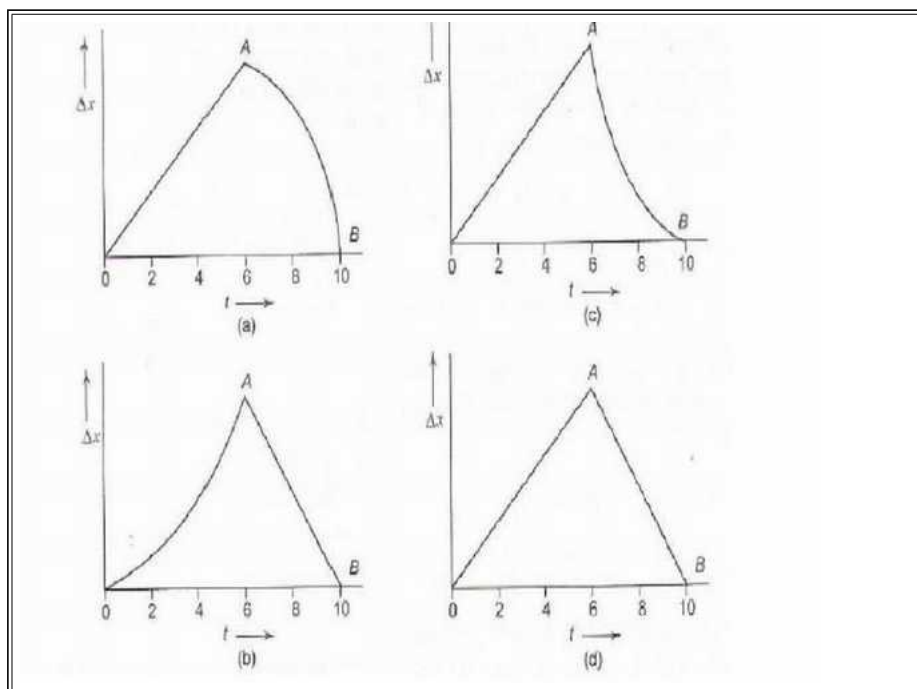
Example. Which of the displacement-time ($x-t$) graphs shown in Fig. can possibly represent one dimensional motion of a particle?



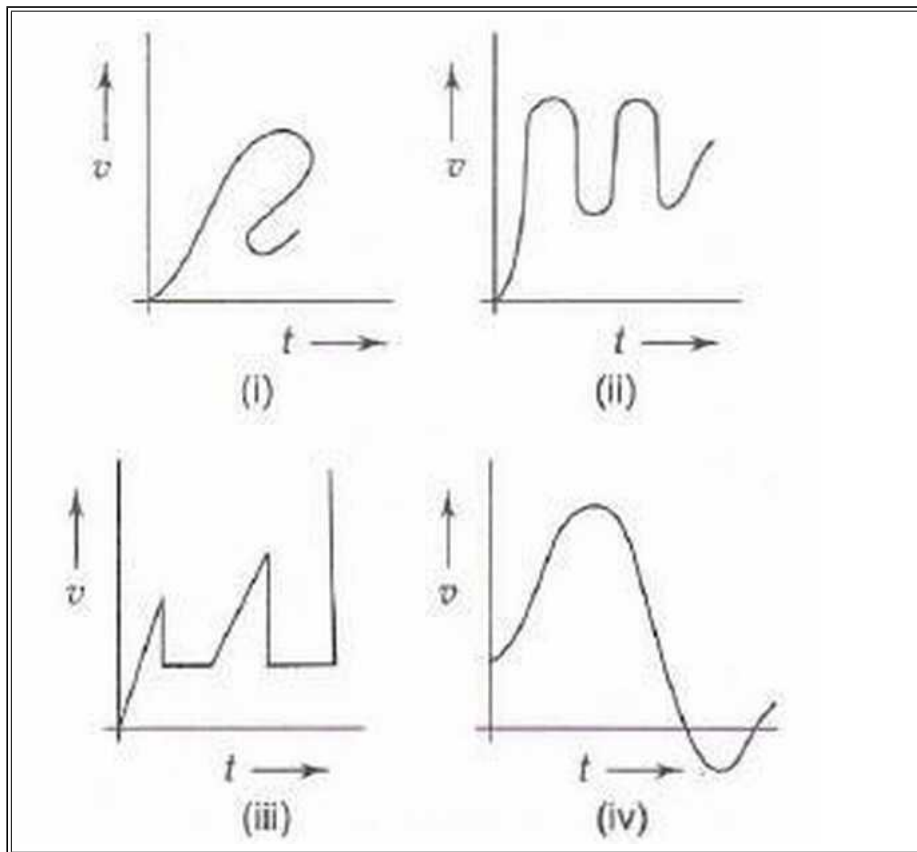
Example. A car starts from rest, accelerates uniformly for 4 seconds and then moves with uniform velocity. Which of the ($x-t$) graphs shown in Fig. represents the motion of the car upto $t = 7$ s?



Example. Two stones are thrown up simultaneously with initial speeds of u_1 and u_2 , ($u_2 > u_1$). They hit the ground after 6 s and 10 s respectively. Which graph in Fig. correctly represents the time variation of $\Delta x = x_2 - x_1$, the relative position of the second stone with respect to the first upto $t = 10$ s? Assume that the stones do not rebound after hitting the ground.

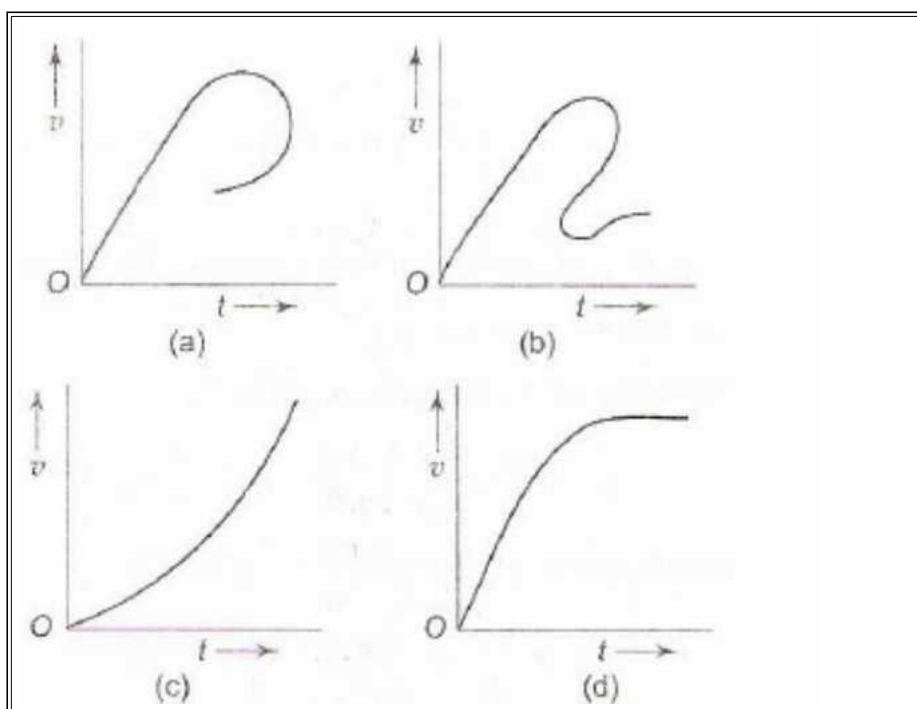


Example. Figure shows the velocity-time ($v - t$) graphs for one dimensional motion. But only some of these can be realized in practice. These are



- a) (i), (ii) and (iv) only
- b) (i), (ii) and (iii) only
- c) (ii) and (iv) only
- d) all

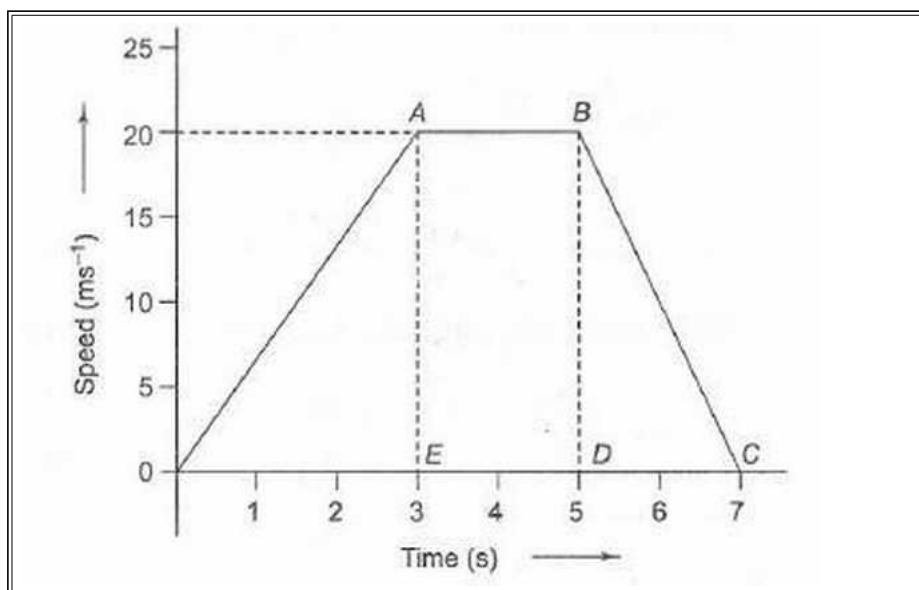
Example. Which of the velocity-time (v - t) graphs shown in Fig. can possibly represent one-dimensional motion of a particle?



1.3 Question Types

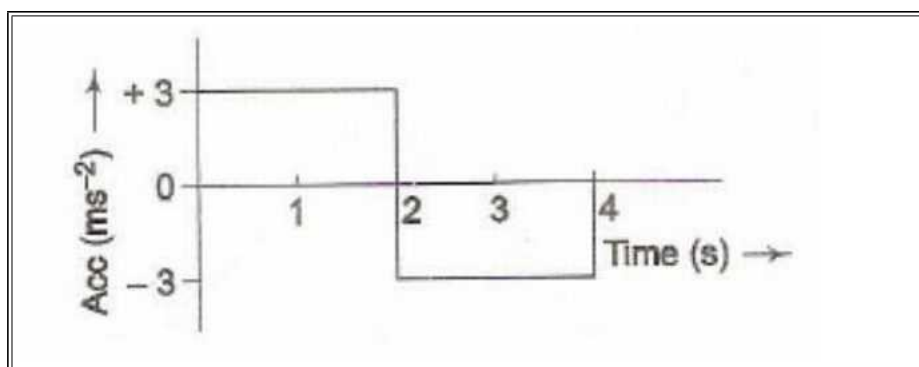
1.3.1 Passage Type

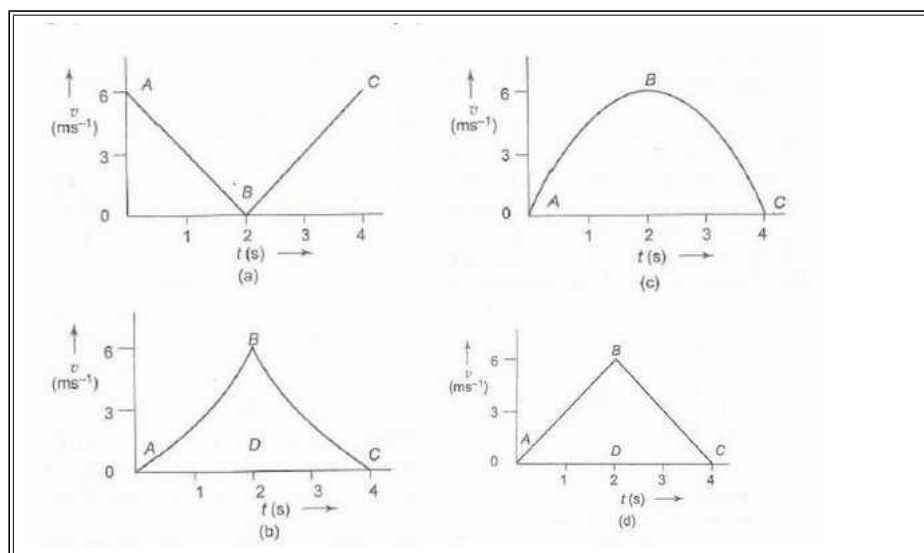
Example: The speed-time graph of the motion of a body is shown in Fig.



- The accelerations of the body during the last 2 second is
 - $\frac{20}{3}ms^{-2}$
 - $-\frac{20}{7}ms^{-2}$
 - $-10ms^{-2}$
 - Zero
- The ratio of distance travelled by the body during the last 2 seconds to the total distance travelled by it is
 - 1/9
 - 2/9
 - 3/9
 - 4/9
- The average speed of the car during the whole journey is
 - 10 m/s
 - 20 m/s
 - $90/7$ m/s
 - $40/7$ m/s

Example: A body starts from rest at time $t = 0$ and undergoes an acceleration as shown in Fig. Which of the graphs shown in Fig. represents the velocity-time ($v-t$) graph of the motion of the body from $t = 0$ s to $t = 4$ s?





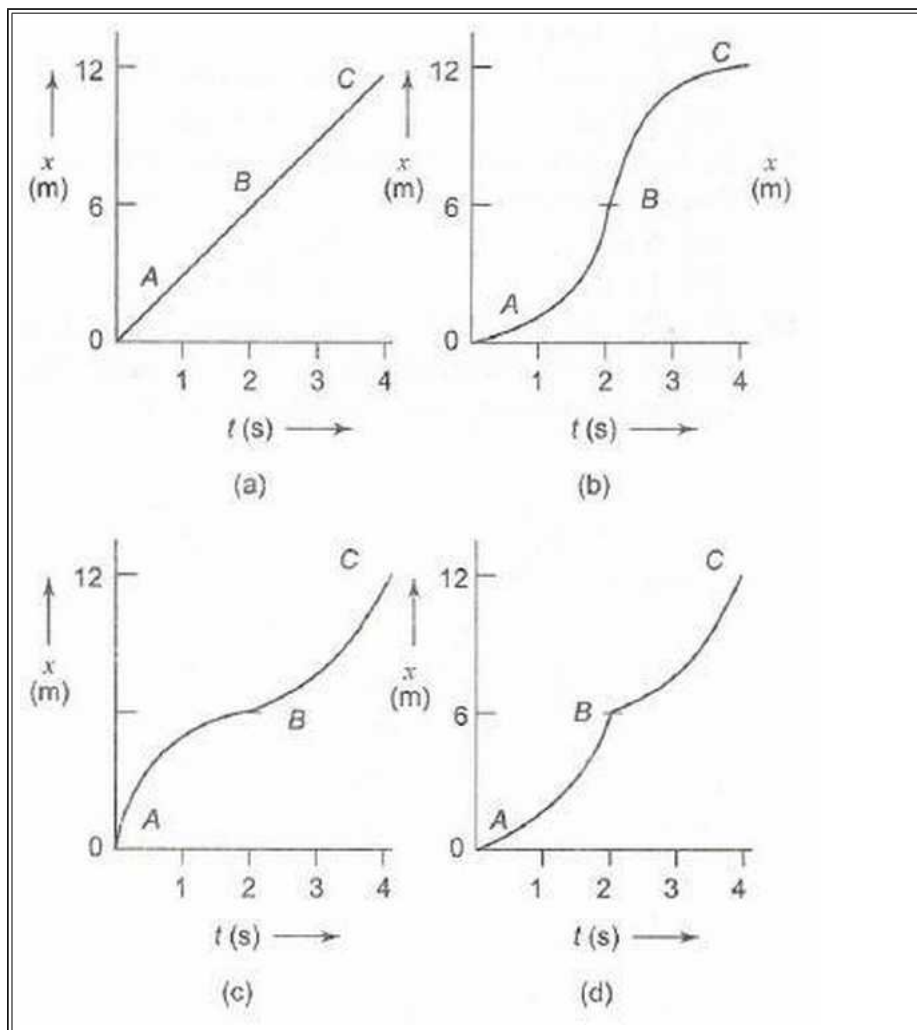
1. In Question above, what is the velocity of the body at time $t = 2.5$ s?

- a) 2.5 m/s
- b) 3.5 m/s
- c) 4.5 m/s
- d) 5.5 m/s

2. In above question, how much distance does the body cover from $t = 0$ s to $t = 4$ s?

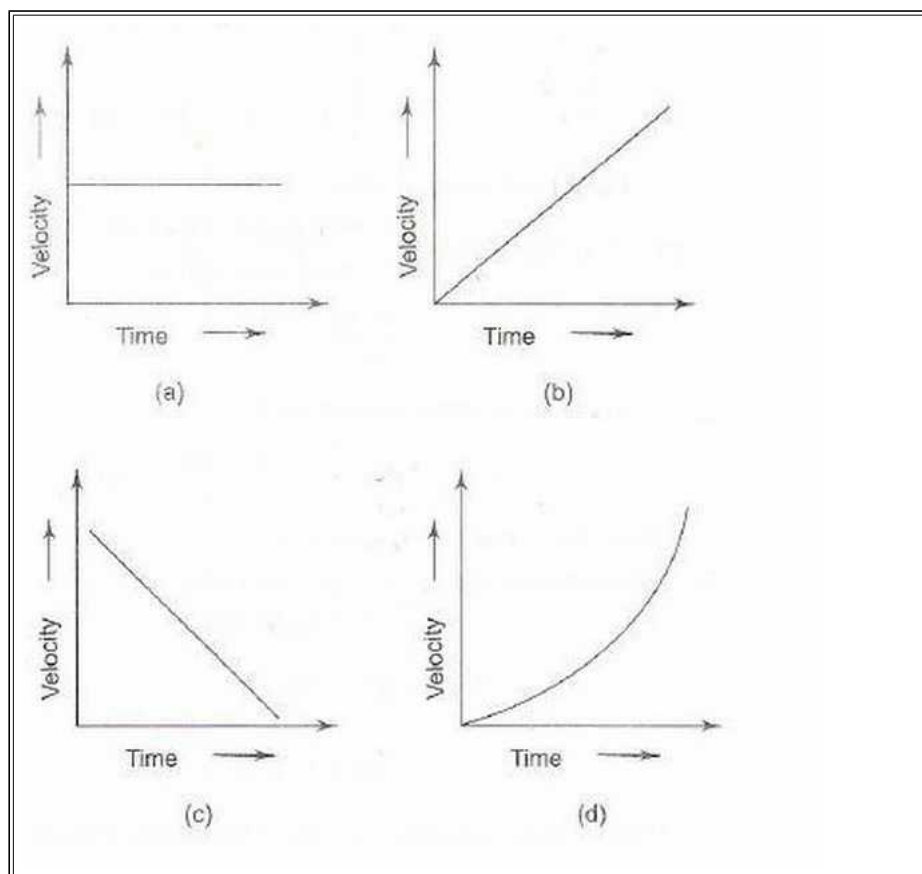
- a) 6 m
- b) 9 m
- c) 12 m
- d) 15 m

3 In above question, which of the graphs shown in Fig. represents the displacement-time (x - t) graph of the motion of the body from $t = 0$ s to $t = 4$ s?



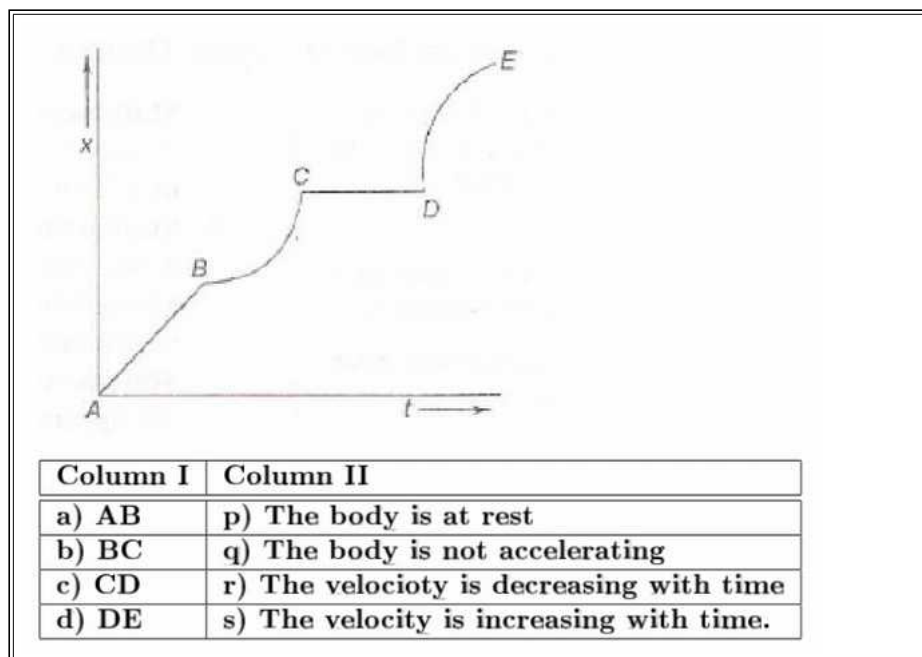
1.3.2 Matching

- Match the graphs (a), (b), (c) and (d) shown in Fig. with the types of motions (p), (q), (r) and (s) that they represent



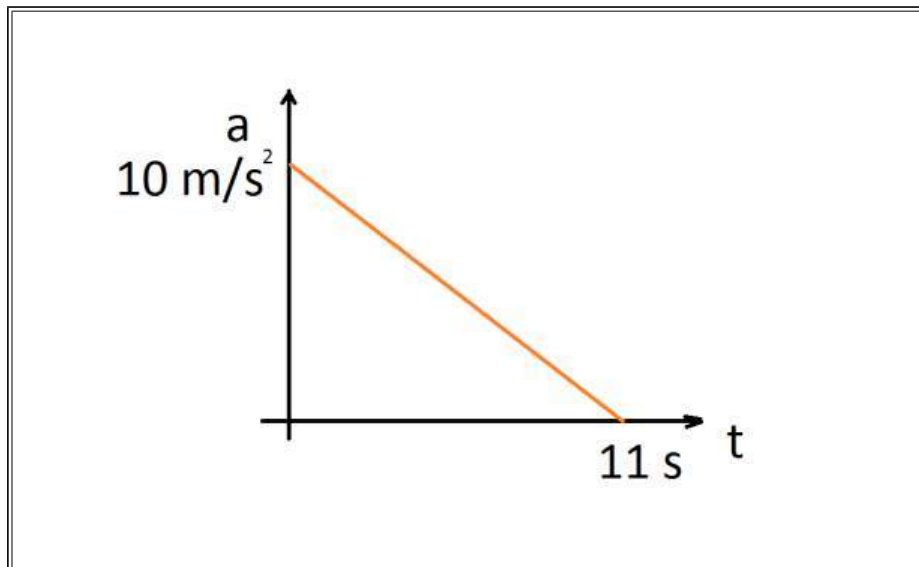
- p) Motion with non-uniform acceleration
- q) Motion of a body covering equal distances in equal intervals of time
- r) Motion having a constant retardation
- s) Uniformly accelerated motion.

2. Figure shows the displacement-time ($x-t$) graph of the motion of a body.



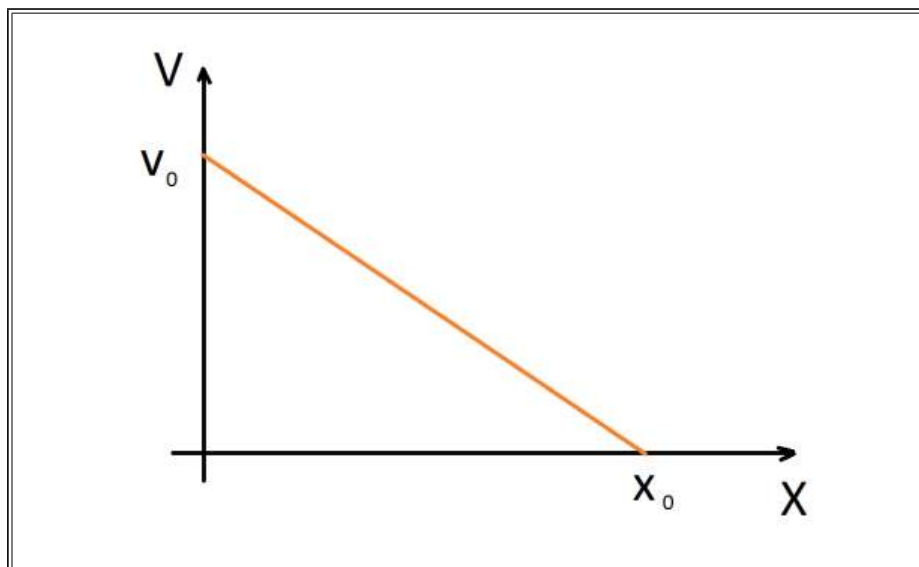
1.4 Previous Year Problems IIT

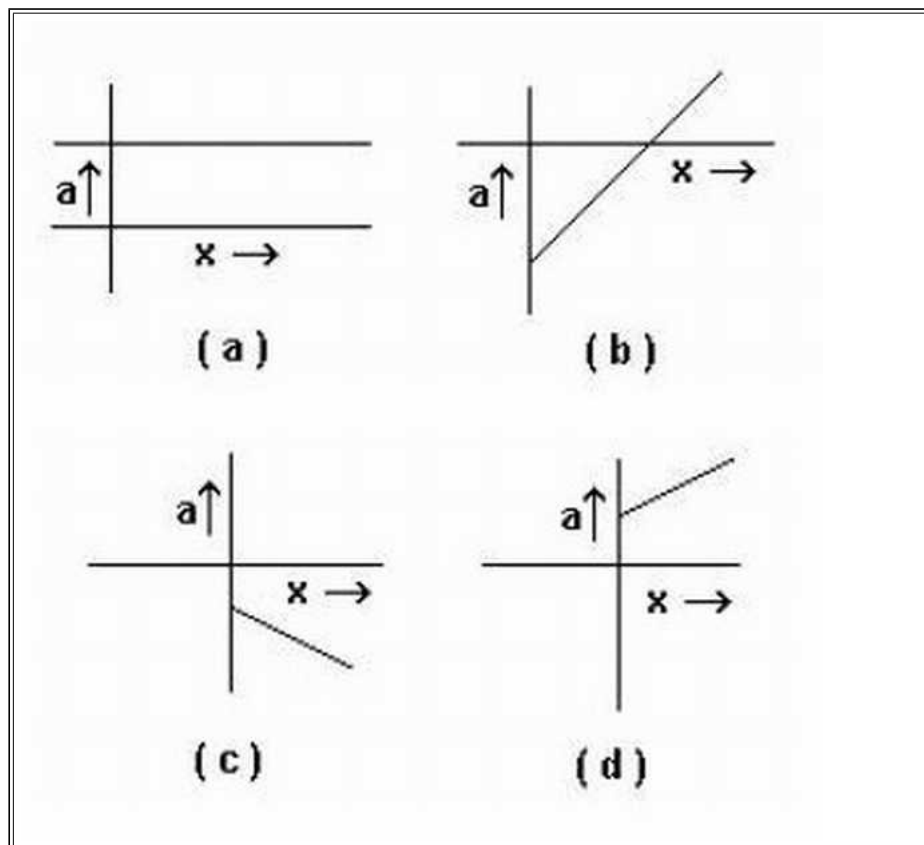
Q1: A particle starts from rest. Its acceleration (a) versus time (t) is as shown in the figure. The maximum speed of the particle will be



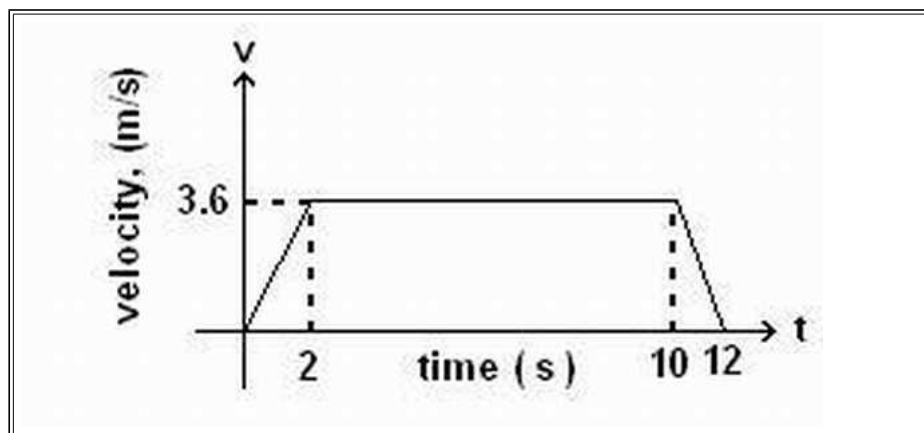
- a) 110 m/s
- b) 55 m/s
- c) 550 m/s
- d) 660 m/s

Q2: If graph of velocity vs. distance is as shown, which of the following graphs correctly represents the variation of acceleration with displacement ?





Passage A lift is going up. The variation in the speed of the lift is as given in the graph.



Q3: What is the height to which the lift takes the passengers ?

- a) 3.6 m
- b) 28.8 m
- c) 36 m
- d) cannot be calculated from the above graph

Q4: In the above graph, what is the average velocity of the lift?

- a) 1 m/s
- b) 2.88 m/s
- c) 3.24 m/s
- d) 3 m/s

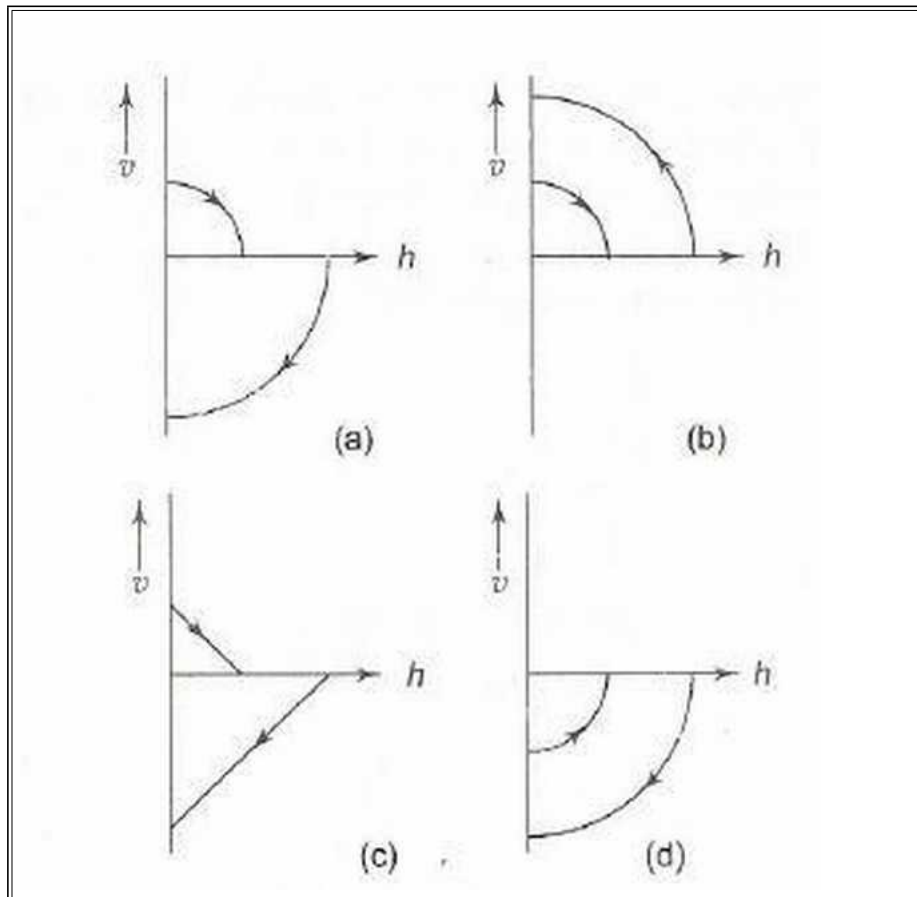
Q5: In the above graph , what is the average acceleration of the lift?

- a) $1.8m/s^2$
- b) $-1.8m/s^2$
- c) $0.3m/s^2$
- d) zero

Q6: Four persons K, L, M and N are initially at the corners of a square of side of length d . If every person starts moving with velocity v such that K is always headed towards L, L towards M, M towards N and N towards K, then the four persons will meet after

- a) d/v s
- b) d^2/v s
- c) $d/2v$ s
- d) $d/2v$ s

Example. A ball is dropped vertically from a height h above the ground. It hits the ground and bounces up vertically to a height $h/2$. Neglecting subsequent motion and air resistance, its velocity v varies with the height h as (see Fig.) (I.I.T. 2000)



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